SPECIFICATION SHEET

FOR

Rapid Roll up Doors

RFQ - E80000119 -005 - R0

PROJECT NO. E80000119

Godrej | CONSOVEYO	Rapid Roll - up Door	RFQ-E80000119-005-R0	
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		Date	07.05.2020
		Page No.	

REVISIONS

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00	First Issue	07.05.2020	GCLA

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1.0 Description:

Rapid Roll - up Doors are doors that covers material on conveyor.

2.0 Requirement

1 Rapid Roll - up Door

2 QTY: 09 no. @ ASRS conveyor area

QTY: 08 no. @ Sampling conveyor area

3.0 Sequence of Operation

1	Carton/Pallet arrive from preceding conveyor to input station of Rapid Roll - up Door.
2	Default position of door will be in CLOSE position i.e. if GCLA command through potential free contact for door open is not available then door will remain in closed position only.
3	Door open command from GCLA system through potential free contacts (NO) will be provided for opening the door. Signal will be steady "ON" during entire operation of carton/ pallet movement beneath the rapid door.
4	Door will get opened as per open command & remain open as long as signal from GCLA is continuously ON for door open.
5	Carton/Pallet will be passed to next conveyor
6	After passing the Carton/Pallet, Door Open command from GCLA system will be made “OFF” resulting in door close operation.
7	Supplier should provide Manual/ Auto mode for equipment (Mechanical & Electrical, Both) using potential free contact.

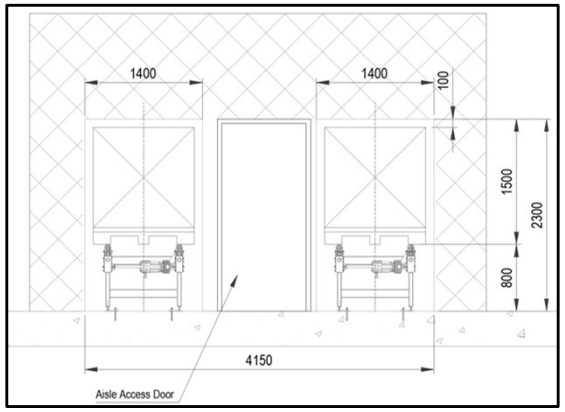
4.0 Technical specification:

Sr. No.	Description	Pallet/ASRS Area	Sampling area
1	Door Opeining Dimensions required	1400(W) x 1500(H)	1000(W) x 1000(H)
2	Component details	Pallets	Cartons/Bags/Drums
3	Component dimensions (Max)	1000mm(W) x 1200mm(D) x 1400mm(H)	550mm(W) x 850mm(D) x 860mm(H)
4	Component Weight (kg)	850	50
5	Type of conveyor	Chain Conveyor	Light Load Roller Conveyor
6	Conveyor Dimensions	960mm(W) x 3000mm(L) x 800mm(H)	960mm(W) x 3000mm(L) x 800mm(H)
7	Height of conveyor	800	800
8	Speed of the conveyor (m/s)	0.3	0.5
9	Accelearation - conveyor (m/s^2)	0.3	0.5
10	No of operation (Carton Per Hour)	40	64
11	No of Shifts	2	2
12	Availability	96%	96%
13	Application	Indoor	Indoor
	Application Requirement		
1	Door construction	Self-supporting	Self-supporting
2	Guide material / surface	PU Painted Galvanised steel	PU Painted Galvanised steel
3	Curtain MOC	1.2mm thick UV protected PVC coated reinforced polyester flexible curtain Class 2, DIN 4102 in self extinguishing material protected against microbial & fungal attack, divided into 3 to 4 sections by extruded aluminium anodised profiles. 900mm vision field in transparent PVC 2mm thick.	
4	Curtain colour	RAL 9002 or Similar	RAL 9002 or Similar
5	Wind load	EN 12424, Class 2	EN 12424, Class 2
6	Opening Speed (m/s)	2	2
7	Closing Speed (m/s)	0.5	0.5

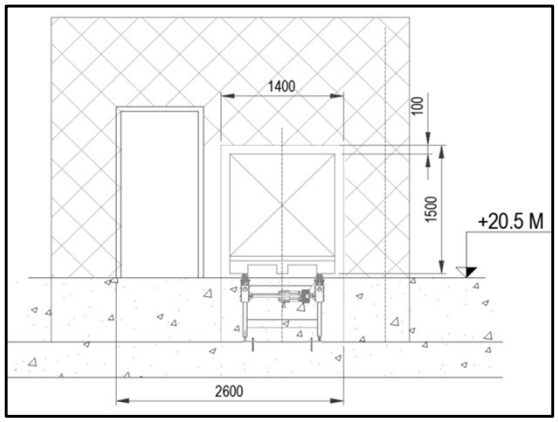
5.0 G A Layout

Please refer attached drawing no.: **RFQ_DRG_E80000119_05 R0 Rapid Roll up Door**

Rapid Roll-up Door drawing for Pallet Conveyor (For reference only)

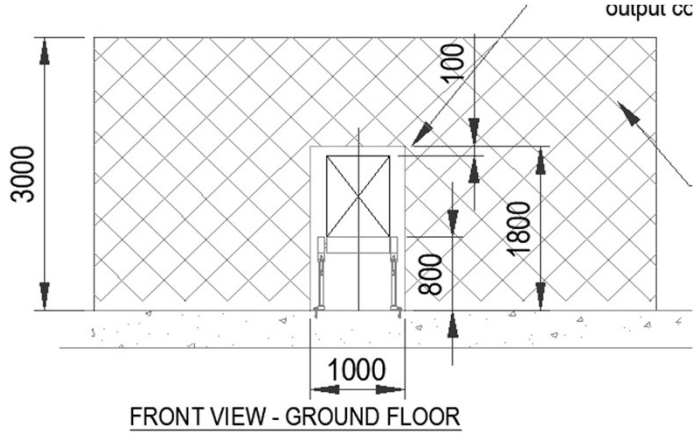


Ground Floor - Pallet Conveyor (For reference only)

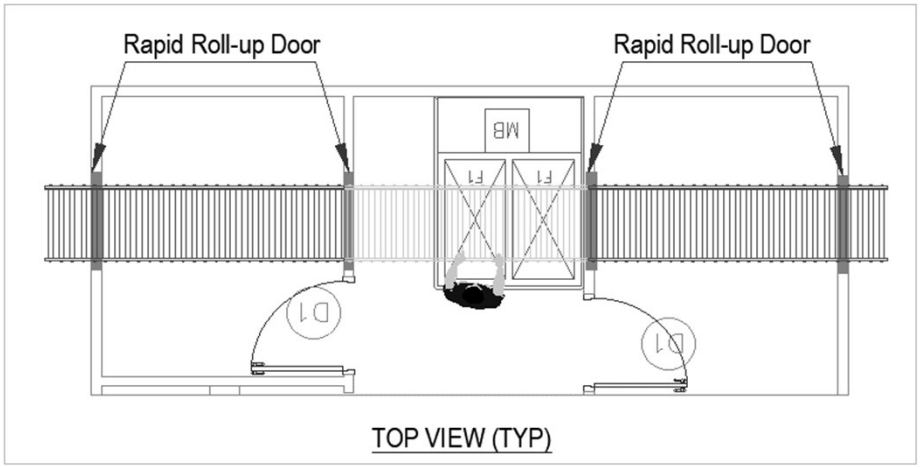


Third Floor - Pallet Conveyor (For reference only)

Rapid Roll-up Door drawing for Carton Conveyor (For reference only)



Front View - Sampling area



Top View - Sampling area

6.0 Reference Images - Rapid Roll - up Door

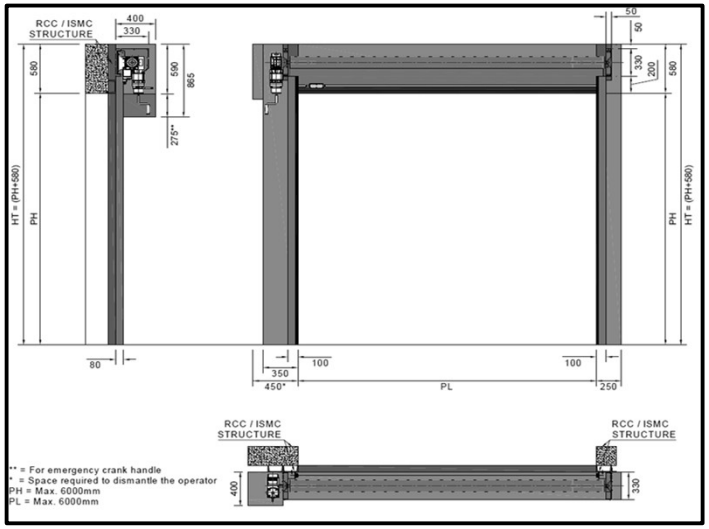


Image - 01. - Mounting Arrangement

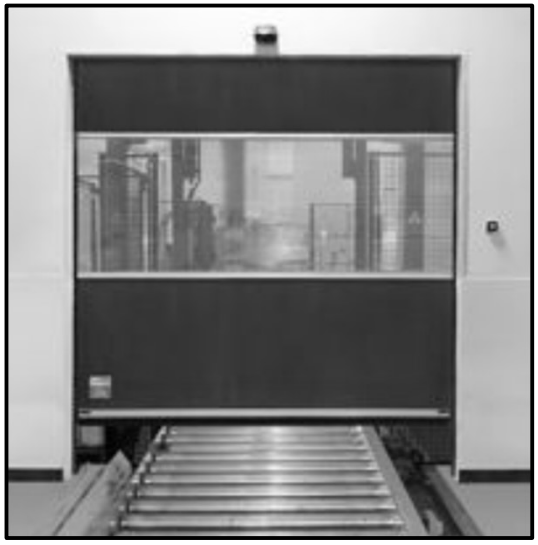


Image - 02. - Rapid roll up door

7.0 Scope of supply

1. Rapid Roll - up Door (6 nos at ASRS ground area, 3 nos at ASRS 3rd floor area & 8 nos at Sampling area (Ground floor)
2. Supervision for installation and signed off.
3. Galvanized Steel casing for top roller assembly
4. Galvanized Steel motor guard
5. Direct drive operator with electromagnetic brake
6. Inverter motion system to adjust different speeds, with progressive control of acceleration and deceleration during braking
7. Operator Panel
8. Advance control device with integration with GCLA PLC system, Consider Potential free contact - 05 Nos. each.
9. All required safety feature and accessories as per industry standard
10. All anchoring fasteners for door & panel mounting

11 Mounting arrangement for Cable drag chain. (If any)

12 All sensors & limit switches including safety switches / devices

13 Cable routing & trays wherever required

14 All sensors & devices termination to be done at supplier factory & to be tested jointly

15 All sensors & devices should be connector type, screw connection / soldering is not allowed.

16 Commissioning spares, Consumables & wear parts

17 Integrated Performance demo, PAC & FAC.

18 Whatever is not included in GCLA scope.

19 Flashing Amber light.

8.0 Safety & interlock requirement

1 Jerk free movement

2 Seals to avoid temperature & noise transfer

3 Should equipped with Self testing safety edge

4 Motor should be guarded propoerly

5 Electromagnetic brake should be included for smooth operation

6 Crank handle for manual operation

7 Auto Door open functionality if accidentally object is detected in shutter path while door closure movement.

8 Shutter door status signal as READY & ERROR to GCLA system via Potential free contact.

9.0 GCLA Scope

1 Sensors for Carton/pallet start stop

2 PLC Programming for GCLA conveyors

3 Support structure as per drawing no 01, on page no 08

4 Power supply to shutter door, 1-phase+E+N @230vAC (+-10%) @50Hz (+-5%)

10.0 Equipment Finish requirements

10.1 Fabrication

1 All spatters to be removed.

2 Welding should be uniform.

3 No blow holes in welding.

4 All critical welds as identified by GCLA shall be checked by DP test.

10.2 Machining

1 All machined surfaces shall be protected with corrosion protective coating.

2 Remove all sharp edges.

10.3 Painting

Sr No	Description	Details
1	Process	Spray painting
2	Paint base	Epoxy - One coat of primer +2 coats of finished paints
3	Cleaning process	Sand blasting as per IS 1477
4	Finishing	Glossy finish
5	Paint thickness	120 - 150 microns

10.4 Plating

Sr No	Description	Details
1	Process	Trivalent chromate plating
2	Thickness	10 - 20 microns

10.5 Powder coating

Sr No	Description	Details
1	Process	Powder coating
2	Paint base	Epoxy
3	Cleaning process	7 tank process
4	Finishing	Glossy finish
5	Paint thickness	60 - 80 microns

10.6 Colours

1 Stationary structures: RAL 5003

2 Moving parts: RAL 1028

10.7 Electrical Wiring

1. All wiring should in compliance with IEC standards
2. Multicore cables with Wires numbering pattern to be used

10.8 IS Standards to be Followed

Sr. No.	Standard	Details
1	IS 800 – 2007	General Construction on steel
2	IS 808 – 1989	Dimensions for hot rolled steel Beam, Column, Channels & angle sections
3	IS 3655 – 1966	Recommended practice for electroplating
4	IS 10748 – 1984	Hot-rolled Steel Strip for Welded Tubes and Pipes - Specification
5	IS 101 – 1964	Methods of Sampling and test for paints, varnishes and related products
6	IS 2074 – 1992	Ready Mixed Paint, Air Drying, Red Oxide Zinc Chrome, Priming - Specification
7	IS 513	Cold Rolled Low Carbon Steel Sheets & Strips
8	IS 325 – 3	Phase induction motor-Specifications.
9	IS 1363	Hexagon Head Bolts, Screws and Nuts
10	IS 2062 – 1999	Steel for General Structural Purposes – Specification
11	IS 4923 – 1997	Hollow Steel Sections for structural use.
12	IS 2016	Specification for plain washer.

11.0 Site Address

GIDC Savli Vadodara 391775, Gujarat

12.0 Site Location & Climatological data

Sr. No.	Description of Parameter	Value	
1	Ambient temperature ^{0}C	Max	45
		Min	5
2	Design temperature ^{0}C	Min 45	
3	Relative Humidity	Max	95%
		Min	11%
4	Wet bulb temperature	Max	30
		Min	5
5	Environment	Dusty	
6	Rain fall, Avg annual mm		
7	Wind, max velocity m/s		
8	Earthquake	As per IS	
9	Plant side elevation above MSL		
10	Topography at site	Flat	
11	Drainage pattern		
12	Nearest highway	NH 48	GJ SH 5
13	Nearest village/town/city		
14	Vadodara	20 Km	
15	Ahmedabad, Gujrat	111 Km	
16	Mumbai Central	405.3 Km	
17	Seismicity	Zone III as per IS 1893:2002	

13.0 Tentative Delivery schedule

Sr. No	Milestone	Date
1	PO from GCLA	
2	Drawing approval	
3	FAT	
4	Dispatch	
5	Delivery at Site	
6	Installation & Commissioning	
7	Integrated Performance Demo	To be Discussed
8	Training and handover	To be Discussed

14.0 List of Approved makes

Sr No	Description	Make
1	Geared Motors & VFD	Sew Euro drive
2	Sensors	Sick / Leuze / P&F
3	Limit Switches	Siemens / Omron
4	Bearings*	SKF / INA / FAG
5	Chains	Ti Diamond / Rollon
6	Electrical Cables	Igus / Laap
7	Anchor fasteners for grouting	HILTI
8	Fasteners	TVS / Unbrako
9	Wire Rope	Usha Martin
10	Structural steel	TATA/JINDAL/SAIL/ESSAR/ Any international make

Note: All bought out components should be of reputed make. Supplier should take prior approval from GCLA

for make of items not covered in above list.

All critical fasteners should be of 12.9 grade. Other fasteners should be minimum 8.8 grade.

* All bearing shall be both side sealed bearing 2RS / 2Z & lifespan should be minimum 5 years with 99% reliability

15.0 DAP Requirement

Sr. No.	Description	Date	Approver
1	GA Drawing in AutoCAD & PDF.		Mechanical Design
2	Floor load with static, dynamic and uplifting forces/ moments in closed and extended condition		Mechanical Design
3	Utility, power & air requirement		Mechanical Design, Automation Design
4	Power requirement		Automation Design
5	compliance with GCLA RFQ		Mechanical Design, Automation Design
6	Approval on RFQ deviataion if any		Mechanical Design, Automation Design
7	Civil requirement like Grouting details and finish floor level		Mechanical Design
8	Spares parts list with separate price with critical & desirable distribution		Mechanical Design, Automation Design
9	List of consumable items with separate price, if any		Mechanical Design, Automation Design
10	Input/Output/Interface details		Automation Design
11	Electric BOM Including Sensors & devices BOM as per GCLA format		Automation Design
12	Complete Datasheets for all geared motor & drives		Mechanical Design, Automation Design
13	Complete Datasheets for all pneumatic components		Mechanical Design, Automation Design
14	Complete Datasheets for all sensors & devices		Automation Design
15	Project Specific Installation Guidelines document		Mechanical Design, Automation Design, Software Design,Installation & Quality Control
16	Integrated Performance Demo procedure & Acceptance criteria		Mechanical Design, Automation Design
17	Quality Assurance plan. Approval on QAP from GCLA quality team is mandatory before Finalization.		Mechanical Design, Automation Design, Quality Control
18	Test certificates of all boughtout items		Mechanical Design, Automation Design Quality Control

**Note - Dates for DAP submission will be confirmed after discussion between GCLA & Supplier

16.0 Acceptance criteria & performance criteria

16.1 Machine Qualification

Sr No	Machine Qualification	Document/criteria
1	Performance test with all type of SKU's/Pallets	Approved drawing
2	Throughput/cycle time test	For min 64 Cycles for sampling area & 40 pallets/ hour for ASRS area
3	Overall size as per approved drawing	Approved drawing
4	Specification & make validation of bought outs	Approved make list
5	Surface finish, aesthetic	As per specification sheet
6	Completeness of assembly	Approved drawing
7	Special requirement if any, as per approval drawing	Approved drawing
8	Scope of supply as per specification sheet	As per specification sheet
9	Closure of all electrical & mechanical safety hazards	As per applicable norms
10	Noise level	As per applicable norms
11	All error demonstration & training	Test to be performed & Certificate to be issue
12	Test certificates for all bought out items like Motor , chains, sprocket, shaft, etc.	Test Certificates
13	Mechanical Up & Down Movement	60 cycles

16.2 Mechanical functional check list

Sr No	Mechanical functional check list	permissible values
1	Door & conveyor Contact seals	Operation test , Visual
2	No metal to metal contact	Operation test , Visual
3	Structural stability	Operation test , Visual
4	Maintenance access to all bearing, motor, limit switches.	Operation test , Visual
5	Emergency access to motor & junction Box.	Operation test , Visual
6	Lubrication point availability and its access level.	Operation test , Visual
7	Effect of overshoot in case of electrical failure. Position of mechanical stoppers	Operation test , Visual
8	Provision of Safety accessories	Operation test , Visual
9	Rapid Roll - up Door uptime	Above 96%
10	Positioning accuracy	± 5 mm (For min 20 Cycles)

16.3 Electrical functional check list

Sr No	Electrical functional check list	permissible values
1	Current drawn for Electrical motor without load, With working load and with Max load.	As per motor specification
2	IO check	As per IO list & BOM
3	VFD integration and compatibility to system. If any	Operation test , Visual
4	Wiring & Cable routing check	IEC standards, Visual
5	JB mounting arrangement check	Visual
6	Multipin connector mounting & termination check	Visual & As per electrical drawings

17.0 Terms & Condition

1 FAT : All equipment must be offered for FAT to check performance parameters. In case it is not possible to check all parameters in FAT it should be demonstrated at site.

2 Confidential Information: Supplier agree that any information received pursuant to this inquiry shall be treated as confidential information and shall not be disclosed to any third party except as is required by law or legal process.

3 List of Critical Spare Parts: Supplier should provide list of spare parts along with quote which are critical for trouble free functioning of the equipment for minimum period of three years after warrantee period.

4 Spare Parts Availability: Supplier should guarantee the availability of spare parts for the equipment for period of 15years from date of supply.

5 Supplier shall provide training to GCLA engineers on adjustment, maintenance and operation of all components.

6 Material Unloading in the scope of supplier.

7 Labors, technicians & supervision in the scope of supplier.

8 Statutory compliance requirements as per law of land and labour laws in scope of supplier.

9 Site Safety rules & regulations should be followed by supplier.

10 Equipment hiring(like Hydra, Mobile Crane, Forklift) in scope of supplier.

11 Cement padding after levelling of machine in the scope of supplier.

Note: Suppliers has to agreed all the points mentioned in RFQ. If there is any deviation, supplier need to take approval from GCLA before sending the offer

18.0 Abbreviations

GCLA: Godrej Consoveyo logistics Automation Ltd

PLC: Programmable Logic Controller

QAP: Quality Assurance Procedure

BOM: Bill of materail

MOC: Material of Construction

19.0 RFQ Approvals

Mechanical Design	Automation Design	Software Design
Procurement	Quality Department	Installation
Projects Head	Engineering Head	Operations Head